

String Programming Questions

Basic

1. Write a program to count the number of vowels in a string.
2. Reverse a string without using built-in functions.
3. Check whether a string is a palindrome.
4. Count uppercase and lowercase letters in a string.
5. Replace all spaces in a string with _.

Intermediate

6. Find the frequency of each character in a string.
7. Remove duplicate characters from a string.
8. Find the first non-repeating character in a string.
9. Check if two strings are anagrams.
10. Convert "hello world" → "Hello World" (title case without using .title()).

Tricky

11. Find the longest word in a sentence.
12. Compress a string like "aaabbc" → "a3b2c1".
13. Count words, characters, and digits in a string.
14. Rotate a string left by n positions.
15. Find all substrings of a given string.

Set Programming Questions

Basic

1. Create a set and add elements dynamically.
2. Find the union and intersection of two sets.
3. Remove duplicate elements from a list using a set.
4. Check if an element exists in a set.
5. Find the difference between two sets.

Intermediate

6. Find common elements in two lists using sets.
7. Check whether one set is a subset of another.
8. Find symmetric difference of two sets.
9. Count unique elements in a list using a set.
10. Remove all common elements from two sets.

Tricky

11. Find missing numbers from 1 to n using sets.
12. Check if two lists have any common elements.
13. Convert a set of strings into uppercase.
14. Identify unique vowels in a given string using a set.
15. Find elements that appear only once in a list.

Dictionary Programming Questions

Basic

1. Create a dictionary and print all keys and values.
2. Count frequency of each word in a sentence.
3. Merge two dictionaries.
4. Find the length of a dictionary.
5. Check if a key exists in a dictionary.

Intermediate

6. Sort a dictionary by values.
7. Find the key with the maximum value.
8. Remove a key from a dictionary.
9. Convert two lists into a dictionary.
10. Count character frequency using a dictionary.

Tricky

11. Invert a dictionary (swap keys and values).
12. Group elements by frequency using a dictionary.
13. Find duplicate values in a dictionary.
14. Create a nested dictionary for student records.
15. Flatten a nested dictionary.

Mixed (String + Set + Dictionary)

1. Count unique words in a sentence.
2. Find common characters between two strings.
3. Find the most frequent character in a string.
4. Remove duplicate words from a sentence.
5. Find words with the same letters (anagram groups).